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Synthesis and Structural-Functional Characterization of Silicon Dioxide from Biowaste Using Acid Extraction

Alibek Mutushev,[1,2,*] Assiya Nuraly,[1,2,*] Aknur Seisenova,[1] Omirzak Kapizov,[1] Dauren Mukhanov,[1,3] Zarina Inelova,[4,*] Zura Yessimsiitova,[1,4] Akzhunis Akimbayeva,[1] Aina Rysmagambetova[5,*] and Mengtay Aitzhan[1,6,*]

Abstract

In recent years, there has been rapid development in acid extraction methods aimed at improving the purity of silicon. In particular, inorganic acids and alkalis are widely used to effectively remove impurities and increase the purity of silicon dioxide (SiO_2), which paves the way for the production of high-quality silicon suitable for use in the electronics industry. This article describes a method for synthesizing silicon dioxide from carbonized rice husks. A comprehensive analysis of the resulting material was conducted using electron microscopy and specific surface area measurements, and the dependence of silicon yield on calcination temperature was investigated. It was found that silicon dioxide-based materials are characterized by a highly developed porous structure and increased electrical conductivity, which allows for a more accurate identification of the relationship between structural features and the structural and physicochemical properties of the electrodes. The results obtained confirm the potential of using biowaste as a raw material base for producing silicon containing materials with specified functional properties. Material characterization was done through scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), and Brunauer–Emmett–Teller (BET) surface area analysis. The effects of the calcination temperature on SiO_2 yield and textural properties were systematically examined. The obtained amorphous silica exhibited a well-developed mesoporous structure with a specific surface area of 120–150 m^2/g. These structural characteristics indicate the potential applicability of the material as a precursor for silicon-based composite anodes. These results provide evidence that rice husk biowaste will be a sustainable and scalable raw material for producing functional silicon-based materials for energy storage applications.

Keywords: Silicon dioxide; High-temperature carbonization; Calcination of carbon–silicon material; Waste recycling; Demineralization.

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1. Introduction

Plant-derived raw materials are increasingly recognized as promising precursors for the development of energy storage materials within the framework of green energy technologies, where the priority is to replace conventional environmentally hazardous inputs with renewable alternatives. In the context of the global energy transition, lithium-ion batteries remain central to electric mobility and renewable energy storage systems; however, their life cycle is accompanied by substantial environmental and technological constraints, which intensifies the demand for sustainable materials and processing strategies.[1-4] Among biomass-derived carbonaceous materials, charcoal and carbonized rice husks are of particular interest due to their availability, low cost, and highly developed porous structure, which confer valuable adsorption and electrochemical properties. Activated carbon is widely applied in lithium-ion capacitors, where charge storage proceeds *via* the electric double-layer mechanism, providing high power density and long operational life. Rice husks are among the most abundant agricultural residues, constituting up to 20% of the harvested rice mass,[5-7] and contain 17-18 wt.% biogenic silica in the form of hydrated Amorphous silicon dioxide (ASD) localized predominantly in the surface layers.

[1]*Scientific Center for New Technologies, 191 Abay Avenue, Almaty, 050046, Kazakhstan*

[2]*Department of Chemical and Pharmaceutical Production Technologies, Faculty of Biotechnology and Chemical Technologies, Almaty Technological University, 100 Tole Bi, Almaty, 050012, Kazakhstan*

[3]*Department of Chemical Physics and Materials Science, Faculty of Chemistry and Chemical Technology, Farabi University, 71 Al-Farabi Avenue, Almaty, 050040, Kazakhstan*

The intrinsic structural characteristics of rice-husk-derived silica make this biomass a highly promising renewable precursor for amorphous silicon dioxide production.[1-3,8] In contrast to conventional silica synthesis from mineral raw materials, which typically requires high temperatures and energy-intensive operations, rice husk processing can be conducted under comparatively mild conditions, thereby reducing energy demand and aligning with the principles of green chemistry and the circular economy.[9,10] The physicochemical properties of the resulting silica are strongly governed by the pretreatment and thermal treatment regimes, including carbonization, calcination, and chemical purification, which determine morphology, porosity, particle-size distribution, and surface chemistry.[11]. During thermal conversion, hydrated amorphous silica bound to the organic matrix undergoes dehydroxylation, silanol condensation, and residual carbon removal, resulting in the formation of thermally stable amorphous SiO_2 with tailored properties. In the present study, these transformation processes were examined using the Kazakhstani rice variety "Marzhan" under semi-industrial carbonization conditions, together with comparative characterization against wood charcoal, demonstrating rice husks as a dual-source precursor of both a conductive carbon matrix and biogenic silica.[12-16]

2. Materials and methods

2.1 Preparation of charcoal and carbonized rice husk samples

Sampling was conducted in accordance with the requirements of GOST 23083–2023 to ensure the representativeness of the test material. Spot samples were collected in containers (buckets, bags), forming a composite sample with a total mass of up to 8 kg. Char samples underwent multistage mechanical processing: in the first stage, the material was crushed to a particle size of 2.8–3 mm; subsequently, for elemental analysis, the material was further ground to a powdery state (particle size less than 60 μm), ensuring a high degree of homogenization. The prepared samples were dried to an airdry state in a drying oven at a temperature of 50–60 °C until a constant mass was reached, which allowed the influence of moisture to be eliminated while preserving the volatile components. After drying, the samples were thoroughly mixed and reduced by quantification to obtain an analytical sample weighing 50–100 g. A vibrating sieve was used to standardize the particle size distribution. The resulting finely dispersed powders of wood charcoal and carbonized rice husks were placed in airtight dark glass containers to prevent moisture sorption from the environment. The samples prepared were used for subsequent analysis by electron microscopy.

2.2 Scanning electron microscopy (SEM)

The morphology and elemental composition of the samples were analyzed at the "National Open-Access Nanotechnology Laboratory" of Al-Farabi Kazakh National University using a Quanta 200i 3D scanning electron microscope (FEI Company, USA). Prior to analysis, the samples were dried; a sample weighing approximately 3 g was placed in the chamber of the instrument's electron-optical system. The microscope is equipped with secondary electron detectors (SED, LF-GSED, GSED) operating in various vacuum modes, which ensure flexibility in the study when the pressure in the chamber changes. Additionally, a backscattered electron detector and a STEM detector were used to obtain images in transmission mode. Image formation was achieved by sequentially scanning the sample surface with an electron beam, followed by the conversion of interaction signals into a visual image. Signal detection was provided by appropriate detectors that convert radiation into an electrical signal, which, after amplification, was used to modulate the screen brightness. For local elemental analysis, energy-dispersive X-ray spectroscopy (EDS, EDAX) with an energy resolution of up to 1000 eV was used, equipped with a semiconductor detector (polymer window, d = 0.3 mm).

2.3 Extraction of silicon from rice husks

Rice husks of the "Marzhan" variety, produced in Kazakhstan, were used as the raw material. The material was first subjected to heat treatment (carbonization). The husks were loaded into 50-liter metal containers, labeled in accordance with the requirements for storage and quality control of raw materials. Carbonization was carried out under industrial conditions in an isothermal mode. The temperature range of the process was 5–350 °C at a relative humidity of no more than 80% (at 25 °C). A forced ventilation system was used to remove gaseous decomposition products. Particular attention was paid to preventing the ingress of foreign solid inclusions (metal, glass, mineral particles) capable of damaging the equipment. The resulting carbonized rice husks were subjected to demineralization. Rice husk ash was obtained by pyrolysis at a temperature of approximately 800 °C. The material was then treated with a 7% solution of nitric acid (HNO_3) for 1 hour to remove residual organic compounds and mineral impurities. After acid treatment, the samples were repeatedly washed with distilled water until a neutral pH (≈7) was reached. Drying was carried out in two stages: at room temperature for 24 hours, then in a drying oven at 70 °C for 12 hours. The final heat treatment (calcination) was carried out in a muffle furnace in the temperature range of 700–1100 °C. The process of

[4]*Faculty of Biology and Biotechnology, Farabi University, 71 Al-Farabi Avenue, Almaty, 050040, Kazakhstan*
[5]*Faculty of Geography and Environmental Sciences, Farabi University, 71 Al-Farabi Avenue, Almaty, 050040, Kazakhstan*
6Department of Fundamental Medicine, Faculty of Medicine and Healthcare, Farabi University, 96 Tole bi street, Almaty, 050012, Kazakhstan
[*]*Email:* ali.mutushev@gmail.com (A. Mutushev), assiya488@gmail.com (A. Nuraly), z.inelova2015@gmail.com (Z. Inelova), Ayna.Rysmagambetova@kaznu.edu.kz (A. Rysmagambetova), mentay1000@gmail.com (M. Aitzhan)

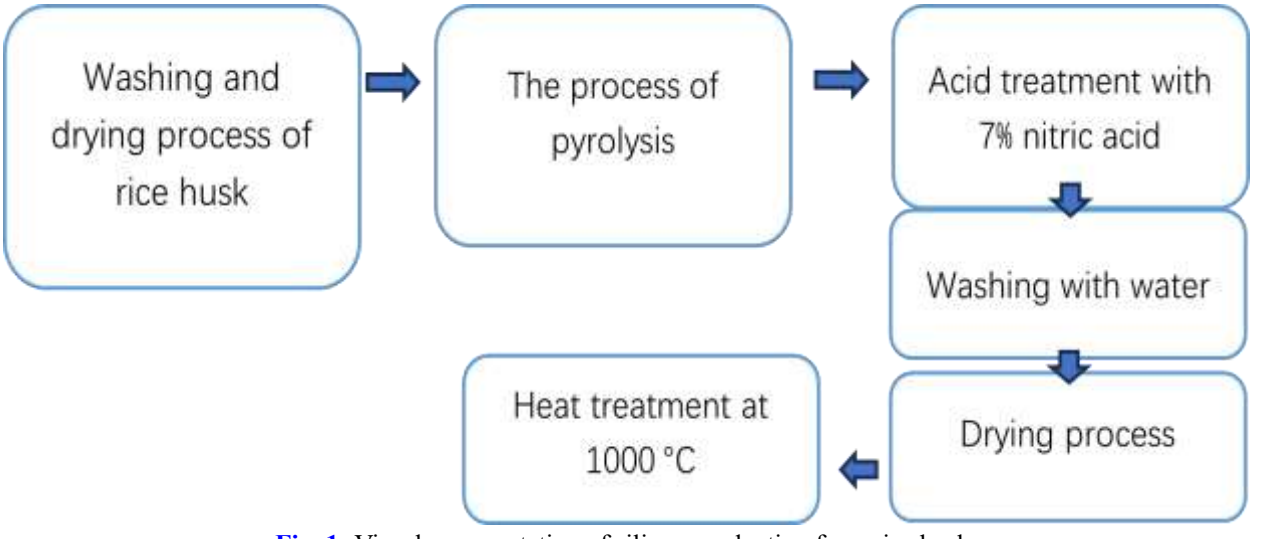


Fig. 1: Visual representation of silicon production from rice husk.

obtaining silicon dioxide from rice husks is a multi-stage process and is determined by a number of factors, among which the temperature of the heat treatment is key. Controlling the temperature regime allows for the regulation of the formation of the amorphous or crystalline phase of SiO_2. The resulting ash contains predominantly amorphous silicon dioxide, suitable for preparation of high-purity silica materials.

To produce high-quality silicon dioxide, it is necessary to strictly control the combustion temperature within a specified range. The production of silicon dioxide from rice husks is a multi-stage process that involves burning the husks to form ash, followed by further processing to extract silica.[13] The ash formed during the thermal processing of rice husks contains predominantly amorphous silicon dioxide (SiO_2), which can be further used to produce high-purity silicon dioxide or reduced to metallic silicon (Fig. 1).

The phase composition and crystallinity of the synthesized SiO_2 were determined by X-ray diffraction (Cu Kα radiation, λ = 1.5406 Å) through a 2θ range between 10 and 80°. The amorphous nature of the silica obtained was confirmed by the analysis.

2.4 Determination of specific surface area (BET method)

The specific surface area of the synthesized silica was determined using the Brunauer–Emmett–Teller (BET) method with a NOVA e-Series analyzer. The instrument's gas system ensured the delivery of the adsorbate gas and carrier gas through a series of purification traps, including liquid nitrogen cooling to remove moisture and volatile impurities. The gas mixture of the desired composition was formed by controlled mixing of the components using flow controllers. The sample was placed in a measuring cell installed in a thermostatically controlled adsorber. The instrument's design allowed for automatic cooling cycles down to the temperature of liquid nitrogen and subsequent heating to the desorption temperature. In the adsorption mode, the sample was cooled to low temperatures, ensuring physical adsorption of the gas onto the surface, whereas during desorption, the adsorbed gas was gradually removed while the pressure change was recorded. Pressure, temperature, and liquid nitrogen level were monitored using built-in sensors and thermocouples, ensuring high measurement accuracy. The obtained adsorption–desorption isotherms were used to calculate the specific surface area, pore volume, and characteristics of the porous structure of the materials under study.

3. Results

The research conducted was motivated by the need to develop effective and environmentally friendly technologies for processing silicon-containing plant waste to produce functional materials with specified properties. Of particular interest are the processes of forming carbon- and silicon-containing structures during the thermal treatment of biomass, as they determine the morphology, phase composition, and performance characteristics of the final products. This article presents the results of a study of the elemental composition, structural and morphological features, and phase transformations of wood charcoal and carbonized rice husks at various stages of thermal treatment. Particular attention is paid to the patterns of silica phase formation and its influence on the properties of the resulting materials. As can be seen, electron microscopies of wood charcoal and carbonized rice husk samples reveals the characteristics of their morphological structure (Fig. 2(a, b)).

3.1 Morphology and elemental composition (SEM-EDS)

SEM images showed distinct morphological differences between the samples (Fig. 2(a, b)). Fig. 2a demonstrates the morphology of the charcoal sample that primarily consists of large lamellar and fragmentary particles that show layered structure. The particles are elongated and flat, characterized by high surface roughness, chipping steps, microcracks, and flaking layers. Fig. 2b presents outline of the carbonized rice husks. The particles have an irregular, angular structure along

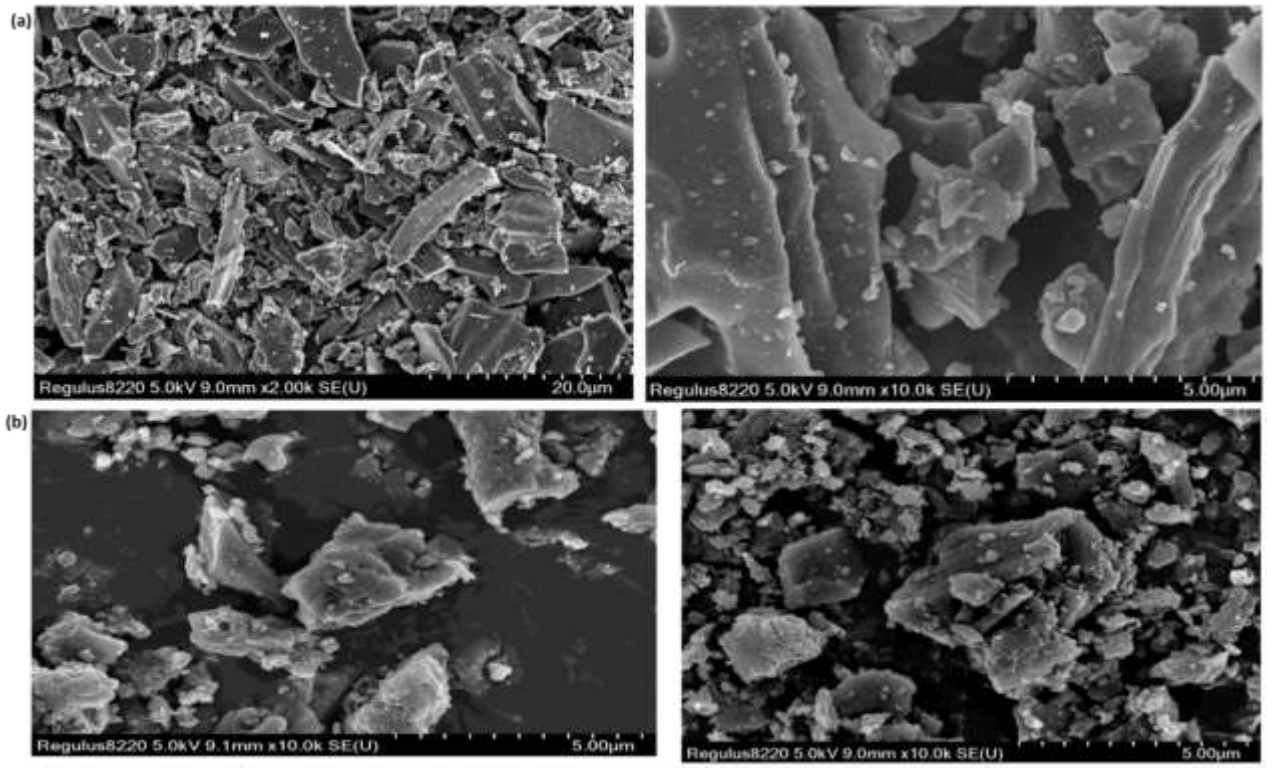


Fig. 2: (a) Morphology of a sample based on charcoal; (b) Morphology of a sample based on carbonized rice husks.

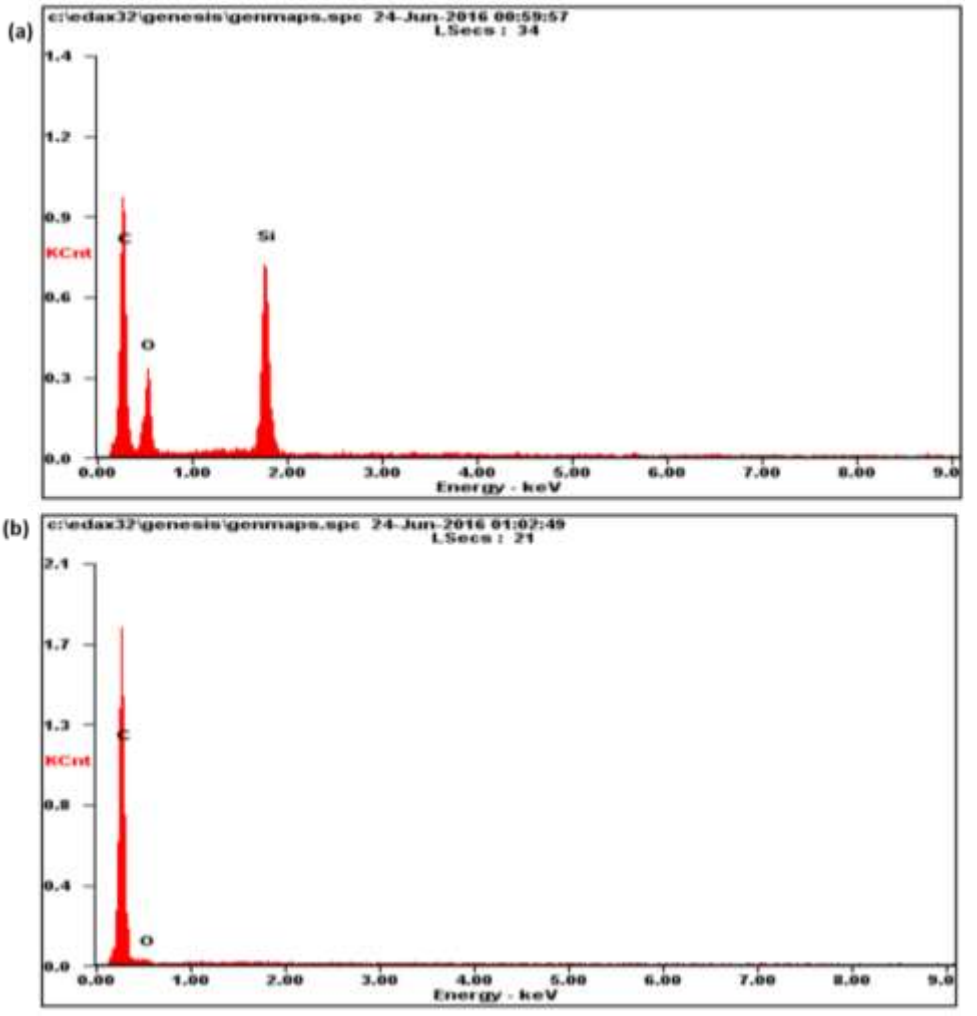


Fig. 3: (a) EDS spectrum of a charcoal sample; (b) EDS spectrum of carbonized rice husks.

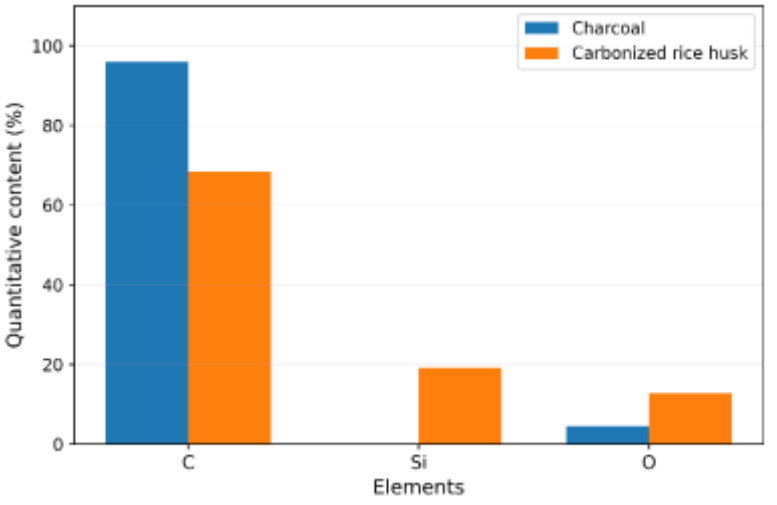


Fig. 4: Quantitative content of elements in samples of charcoal and carbonized rice husks.

with high roughness on the surface, and local porosity. Fine nanoscale formations can be seen on the bigger fragments, suggesting the formation of a hierarchical structure. EDS spectra confirmed the elemental differences (Fig. 3(a, b)). Fig. 3a (charcoal) shows intense peaks for carbon (C) at ~0.27 keV and oxygen (O) at ~0.52 keV with no significant features higher than 1 keV. Fig. 3b (carbonized rice husks) reveals clear peaks for carbon (C), oxygen (O) and silicon (Si) at around ~1.74 keV. The ratio of oxygen to silicon intensity confirms that SiO_2 is present. Fig. 4 shows quantitative elemental composition. Using these results, carbonized rice husks were used as raw material for the further synthesizing of SiO_2.

Fig. 2a shows the morphology of a charcoal sample. Analysis of the image reveals that the material consists primarily of large lamellar and fragmentary particles with a distinct layered structure. The particles are predominantly elongated and flat in shape. Their surface is characterized by high roughness, the presence of numerous chipping steps, microcracks, and fracture zones along the edges. Thin flaking layers and secondary fine fragments localized on the surface are also observed; in some areas, local amorphization of the edge regions is noted.

Fig. 2b shows the morphology of carbonized rice husks. The particles have an irregular, predominantly fragmentary and angular shape, with no signs of crystalline faceting. The surface is characterized by significant roughness and local porosity. On the surface of larger fragments, fine nanoscale formations are observed, indicating the formation of a hierarchical structure in the material. Such morphology contributes to an increase in specific surface area and, consequently, to an increase in the material's reactivity.

Fig. 3a shows the EDS spectrum of a charcoal sample. Each peak in the spectrum corresponds to a specific chemical element. In the low-energy region, characteristic peaks of carbon (C) at ~0.27 keV and oxygen (O) at ~0.52 keV are observed. The main part of the signal is concentrated precisely in this range, indicating a predominance of light elements in the sample's composition. The high intensity of the carbon peak indicates a significant carbon content, whereas the presence of oxygen may be associated with the presence of oxygen-containing functional groups or oxide compounds. The absence of pronounced peaks in the energy range above 1 keV indicates a negligible content or absence of elements such as Si, Fe, Al, and others.

Fig. 3b shows the EDS spectrum of carbonized rice husks. The spectrum clearly shows peaks for carbon (C) at ~0.27 keV, oxygen (O) at ~0.52 keV, and silicon (Si) at ~1.74 keV. The carbon peak is characterized by high intensity, the oxygen peak by moderate intensity, and the silicon peak by pronounced medium/high intensity. The presence of a significant silicon signal indicates its substantial content in the sample. The ratio of oxygen to silicon intensities indicates the likely presence of silicon dioxide (SiO_2). Overall, the obtained spectra confirm that the material consists of a carbon matrix with silica (SiO_2) inclusions, which is a characteristic feature of products from the thermal processing of plant raw materials.

Following the identification of silicon-containing components in carbonized rice husks, this material was selected as the starting material for subsequent experiments aimed at producing silicon dioxide. The obtained results confirmed that carbonized rice husks are an appropriate precursor for silicon dioxide synthesis. The obtained amorphous SiO_2 possesses structural characteristics that make it a promising precursor for the future fabrication of Si-C composite anodes for lithium-ion batteries. During electrochemical operation, partial reduction of SiO_2 to electrochemically active silicon is possible, accompanied by the formation of inert lithium silicate phases as well as the formation of a conductive carbon matrix. This structural-phase combination facilitates effective compensation for volume changes in the material during cycling and, as a result, increases the longevity of lithium-ion batteries.

3.2 Carbonization of biowastes

The properties of samples containing rice husks were experimentally investigated under laboratory conditions. The

overall effect of carbonization temperature on mass loss is presented in Table 1. Mass loss up to 500 °C reaches 29.3%, indicating that the organic components of rice husks undergo intense thermal transformations. The decrease in sample mass occurs gradually with increasing temperature, which corresponds to the decomposition of organic matter and the formation of a porous carbon-silica structure at higher temperatures. During carbonization, thermal behaviour and mass loss are evaluated. Table 1 demonstrates the influence of carbonization temperature on mass loss. Mass loss gradually increased by temperature, at 500 °C at 29.3%, and at 850 °C at 55%, while the carbon content as highest (51.8 wt.%) was observed in the 800–850 °C range, versus 35.4 wt.% in the original dried material.

Table 1: Change in the mass of rice husk samples as a function of carbonization temperature.

Sample	Rice husks					
Carbonation temperature, (°C)	300	350	400	450	500	550
Weight loss, (%)	-	-	21,4	23,7	29,3	32,6
Carbonation temperature, (°C)	600	650	700	750	800	850
Weight loss, (%)	35	35	40%	48	51	55

Elemental analysis conducted to assess the effect of carbonization temperature on carbon content revealed significant changes in composition depending on the temperature regime. The maximum carbon content (51.8 wt.%) was recorded in the 800–850 °C temperature range, whereas in the initial dried material its content was 35.4 wt.%. The results obtained indicate the presence of a multi-step thermochemical mechanism governing the carbonization process, involving sequential stages such as the decomposition of biopolymer components and the formation of a carbon matrix. At low temperatures (100–150 °C), predominantly physical desorption of moisture bound by capillary and physical interactions occurs, without significant structural changes. In this range, the mass decreases by 5–10%, which is due to moisture removal. As the temperature rises to 200–350 °C, intensive thermal decomposition of hemicellulose and cellulose occurs, accompanied by the breaking of C–O and C–C bonds with the release of volatile compounds (H_2O, CO, CO_2, low-molecular-weight organic compounds) and a mass loss of approximately 25–40%. Further heating in the 350–500 °C range promotes the decomposition of lignin, which proceeds over a wider temperature range at a slower rate compared to cellulose. At this stage, a denser carbon structure forms, and the mass decreases by 10–20%. In the 500–700 °C range, the main carbonization stage is completed: residual volatile components are removed, the proportion of fixed carbon increases, and chain aromatization and the ordering of structural elements occur. At temperatures above 700 °C, mass losses are minimal (usually less than 3–7%), indicating the high thermal stability of the resulting carbon product. The remaining mass is primarily due to the inorganic silica phase and the stabilized carbon matrix. The average percentage of mass and energy loss in the 600–800 °C range is 65–75%. Optimal process parameters are determined by a trade-off between the yield of the target product and the molecular structure. To obtain highly thermally stable, durable carbon-silica materials, it is recommended to perform carbonization in the 400–600 °C range, which ensures a balance between structural integrity and process efficiency.

3.3 SiO_2 yield and textural properties after calcination

The yield of silicon in the form of SiO_2 after calcination in a muffle furnace was 30–35 wt.% relative to the mass of carbonized rice husks, which is an indicator of a high degree of conversion of biogenic silica under controlled calcination temperatures. The resulting amorphous silicon dioxide had a specific surface area in the range of 120–150 m^2/g and a color ranging from light gray to pure white, which is due to varying degrees of organic residue removal and condensation of the silica network. The total pore volume (0.175–0.287 cm^3/g) and average pore radius (1.69–2.31 nm) indicate a mesoporous structure formed due to a disordered siloxane (Si–O–Si) framework. At the molecular level, the textural properties are associated with the formation of surface silanol groups (≡Si–OH), which participate in the condensation reaction during calcination. An increase in temperature during calcination intensifies the dehydration-condensation processes of silanol groups, which contributes to the formation of an amorphous Si–O–Si network with characteristic structural disorder. The balance between network condensation and the preservation of defective regions plays a key role in maintaining the material's high specific surface area and pore volume. Preliminary washing of rice husks is an important step that ensures the removal of mechanically bound impurities, water-soluble inorganic compounds, and the partial destruction of the organic matrix. This treated material exhibits an increased silica content and promotes the formation of a homogeneous amorphous SiO_2 network. As a result, the density of surface silanol groups increases, which in turn increases the material's specific surface area. The absence of a washing stage results in the production of dark gray amorphous SiO_2 with a SiO_2 content of less than 92%, a low specific surface area (up to 70 m^2/g), incomplete removal of carbon residues, and poor development of the siloxane network. This condition impairs porosity and reduces the surface reactivity of the matrix. The effect of calcination temperature (700–1100 °C) on silicon content and structural evolution was investigated using electron microscopy. An increase in temperature promotes the decomposition of residual organic matter and the reorganization of the amorphous silica network at the molecular level, which directly affects the physicochemical properties of the resulting SiO_2. The results confirm the existence of a close relationship between the process parameters, structure, and functional characteristics of

amorphous silica obtained from bio-waste.

At the Scientific Center for New Technologies laboratory, the specific surface area of amorphous silicon dioxide samples was determined using a Nova 2000e specific surface area analyzer. Fig. 3a shows a linear plot of the adsorption isotherm, illustrating the dependence of the adsorption capacity on relative pressure. The graph depicts the adsorption process as the pressure ratio P/P0 varies from 0.05 to 0.30. Fig. 3b shows a graph illustrating the relationship between adsorption capacity and pressure. Based on BET measurements of the specific surface area, a graph was plotted showing the dependence of the specific surface area of the samples on firing temperature.

According to the data presented in Fig. 4, the decrease in the specific surface area of silicon dioxide with increasing sintering temperature is due to thermally activated processes of sintering, compaction, and crystallization. These processes lead to a decrease in the material's porosity and an increase in the effective particle size. It is assumed that in the temperature range of 700–900 °C, surface hydroxyl groups are removed, accompanied by a decrease in the concentration of active sites, a reduction in surface micro-roughness, and densification of the structure. At temperatures above 1000 °C, the mobility of silicon and oxygen atoms increases significantly, and volume diffusion mechanisms are activated, leading to a reduction in the number of interparticle boundaries. Simultaneously, the pore walls densify and coalesce, resulting in a further decrease in the specific surface area. The specific surface area of the synthesized silicon dioxide is determined by its finely dispersed structure, degree of porosity, and the presence of surface silanol groups formed during the hydration process. The material's high specific surface area is due to the formation of hydrogen bonds. The surface of silicon dioxide consists of silanol groups and siloxane bridges, which ensure its high chemical activity and ability to adsorb water and organic compounds. During the synthesis process (*e.g.*, in the production of precipitated silicon dioxide or aerosil), thermal oxidation reactions occur, leading to the formation of nanoscale particles with a highly developed porous structure. The elemental composition and morphology of carbonized rice husks were investigated before and after calcination in a muffle furnace at a temperature of 1000 °C. The textural characteristics and elemental distribution in the powdered samples were studied using scanning electron microscopy (SEM). SEM images indicate the formation of a hierarchically organized porous structure comprising micro- and mesopores, suggesting a significant restructuring of the material during heat treatment. Analysis of the pore size distribution showed that their average diameter is about 15 nm, indicating a predominance of a mesoporous structure. The formation of the mesoporous architecture is associated with the thermal decomposition of the organic matrix and the simultaneous formation of a carbon-silica framework in which inorganic silica domains are uniformly distributed within the carbon matrix. At the molecular level, such a structure is characterized by a high number of accessible surface sites associated with carbon defect regions and silanol groups (≡Si–OH). The presence of mesopores ensures effective molecular interaction between carbonized rice husks and the adsorbed substances without significantly altering the material's chemical structure. This is of fundamental importance for adsorption processes, which occur primarily due to physical sorption and weak intermolecular interactions. As a result, the resulting carbon-containing material demonstrates high sorption capacity toward a wide range of substances, including heavy metal ions, gases, alkaloids, toxins, glycosides, as well as various organic and inorganic compounds. The binding mechanism is determined by surface-dependent interactions, driven by pore accessibility and surface chemistry. Calcination at 1000 °C enhances the structural stability and uniformity of the porous network while preserving the mesoporous architecture, which is critical for effective adsorption. The results obtained indicate a direct correlation between heat treatment conditions, morphological evolution, and the surface functionality of materials based on carbonized rice husks.

Elemental analysis of carbonized rice husks prior to calcination in a muffle furnace revealed the presence of carbon, silicon, oxygen, and mineral impurities in all samples examined. Silicon dioxide in the raw material is present in a biogenic form, predominantly as hydrated amorphous silica ($SiO_2 \cdot nH_2O$), localized in the outer layers of the husk. During thermal treatment, a sequential transformation of the chemical state and local environment of silicon atoms in the silicon-containing phase occurs. At a temperature of about 400 °C, silicon is retained in the form of hydrated amorphous silicon dioxide, characterized by a developed system of interactions with the organic matrix due to hydrogen bonds and weak covalent bonds of the Si–O–C type. In this case, the dehydroxylation of the silica surface is incomplete due to the removal of both physically and chemically bound water. The mechanism of thermal decomposition can be conditionally divided into two main stages. In the first stage, the organic component, comprising complex hydrocarbon compounds, undergoes destruction with the release of volatile products (H_2O, CO_2, *etc.*), which contributes to the formation of a well-developed porous structure. In the second stage, the silica phase undergoes transformation: amorphous SiO_2 remains in an inorganic state, while carbon is oxidized by oxygen to form gaseous products. The oxidation of residual carbon is initiated as early as 300–450 °C and is accompanied by the formation of CO and CO_2. The gradual burnout of the carbon matrix leads to the release of the silica phase, previously incorporated into the organocarbon framework. At this stage, silicon is predominantly in the +4 oxidation state and is present as amorphous hydrated silicon dioxide.

In the temperature range of 400–600 °C, a densification of the silica structure is observed, accompanied by the breaking of Si–O–C bonds and the formation of stable siloxane bridges (Si–O–Si). Silanol groups (≡Si–OH) are involved in dehydration-condensation-type reactions, forming a more

tightly bound three-dimensional network, which leads to an increase in the degree of polycondensation of the amorphous structure and a simultaneous decrease in the content of carbon impurities due to the ongoing oxidation of residual carbon. At temperatures above 700 °C, further densification of the silica phase and partial structural ordering occur. However, the transition to crystalline modifications (quartz, cristobalite) generally does not occur without prolonged holding at temperatures above 900–1000 °C. The preservation of the amorphous state is due to the high dispersion of the material, the presence of impurities, and the limited duration of high-temperature exposure. To determine the optimal conditions for obtaining amorphous silicon dioxide with specified structural characteristics, the SiO_2 content was calculated in samples obtained at various calcination temperatures. The developed method for producing an anode material based on silicon dioxide and carbon involves the dispersion of SiO_2 using ultrasonic treatment, followed by mixing with wood-derived carbon and heat treatment in an inert atmosphere, ensuring the formation of a composite material with specified morphology and functional properties.

4. Discussion

The comparison between wood charcoal and carbonized rice husks as biomass precursor implies that the morphological, elemental, and thermal properties, which closely depend on the native source and the biogenic composition of the precursor materials, are essentially dissimilar. These differences highlight the enhanced potential of carbonized rice husks for the sustainable production of silicon-based functional materials that can be used to support energy storage systems, particularly the lithium-ion batteries (Li-ion batteries) and capacitors. Distinct morphological features were identified using scanning electron microscopy between the two materials (Fig. 2(a, b)). Wood charcoal exhibited a mainly lamellar and fragmentary structure with elongated flat particles, large surface roughness, microcracks and layers of flaking (Fig. 2a). This morphology is associated with the carbonization of lignocellulosic biomass, characterizing the preservation of the intact plant cell walls, and producing an even but less hierarchical porosity. Conversely, the organic rice husks carbonized exhibited non-rigid, angular particles with high rough surfaces, small local porosity, and nanoscale formations on larger fragments (Fig. 2b). The hierarchical pattern obtained in rice husk samples reflects the naturally occurring distribution of biogenic silica within the surface layer of rice husk, which is maintained by thermal processing and generates more nanoporous domains. This morphology greatly improves the characteristics of the specific surface area and activity of the material, the carbonized rice husk is probably the promising precursor for composite electrode based on their properties compared to standard charcoal. These structural distinction were confirmed through energy-dispersive X-ray spectroscopy (EDS) that verified silicon in the rice husks (Fig. 3(a, b), Fig. 4). The charcoal samples were dominated by carbon (high-intensity C peak ~0.27 keV) and oxygen (~0.52 keV) with almost invisible contributions from other species in keeping with its importance as a highly purified carbon matrix or conductive electrode enhancer in battery electrodes. Carbonized rice husks on the other hand had well defined peaks for carbon, oxygen and silicon which were around 1.74 keV, this signified high SiO_2 content in carbon matrix. The intensity of O and Si signals indicates that amorphous silica phases have formed as well, which complies with nature of silicon dioxide in rice husks (17–18 wt.% in raw material), and its carbonization and calcination transformation. The silicon composition allows rice husks to act as a 2-source precursor (carbon + SiO_2), rather than wooden charcoal, providing a low-cost and eco-friendly substitute for mineral quartz for silicon development. The carbonization thermal behaviour offered further insight into comparison of process efficiencies and material stability (Table 1). The mass loss of rice husks was quite gradual where from 29.3% at 500 °C to as much as 55% at 850 °C, suggesting linear degradation of hemicellulose, cellulose and lignin. The maximum carbon content (51.8 wt.%) reached at 800-850 °C, in contrast with 35.4 wt.% from the first dried material. The resulting multi-stage processes: desorption (<150 °C), biopolymer (200–350 °C), lignin (350–500 °C), and aromatization (>500 °C), are consistent with known pyrolysis processes in biomass, and establish the carbonization to a controlled temperature of 400–600 °C maximizes the trade-off among carbon yield, structure integrity, and energy demand. At elevated temperatures, the retained inorganic silica phase results in thermal stability of rice husk-derived materials that is otherwise limited by pure charcoal. The overall observations provide a framework for a large scalable low-energy production method for the conversion of rice husks into high-purity amorphous SiO_2 and afterward into SiO_2–C composites (Fig. 1). The process employs acid demineralization, pyrolysis at ~800 °C and calcination (700–1100 °C) to phase-control silica production with dehydroxylation and siloxane condensation, while valorizing organic waste. In contrast to silicon production based on quartz, which requires long-term, energy-intensive high-temperature reduction, the method proposed is performed under milder conditions to decrease the energy requirements and alleviate the environmental problems of rice husk disposal from Kazakhstan and comparable regions. The Si-C composites prepared from carbonized rice husks may serve as promising anode materials in the lithium-ion battery. This carbon matrix can stabilize the volume expansion in the lithiation/delithiation process, and SiO_2 can be partially electrochemically reduced to active silicon to improve specific capacity. This synergistic response is much stronger than the pure performance of charcoal additives, which improves cycling stability and power density in lithium-ion capacitors and hybrid devices. Thus, the work can be viewed as promoting the sustainable transition to clean energy by showing how biogenic raw materials can be substituted for environmental burdens during the battery life. The present

study provides some clear benefits of rice husk precursors from a comparative characterization, but it should be followed by a determination of the specific surface area (BET) and electrochemical behaviour of the newly produced composites. Furthermore, scaling up in an industrial context and evaluating long-run environmental impact will add additional evidence to its contribution to the circular economy.

5. Conclusion

The study established a correlation between the parameters of thermal treatment and the formation of the composition, structure, and morphology of carbon- and silicon-containing materials of plant origin. It was shown that charcoal is predominantly carbon-based, with a minimal content of oxygen-containing functional groups and a negligible proportion of mineral impurities, whereas carbonized rice husks constitute a carbon-mineral composite in which biogenic silicon dioxide is uniformly distributed within the carbon matrix. It has been established that the presence of the silica phase is a determining factor in the structural complexity and functional activity of the material, as well as in the differences in its thermal stability, pore structure, and adsorption characteristics compared to carbon analogues. It has been shown that thermal treatment initiates sequential processes of organic component destruction, oxidative carbon removal, and silica condensation with the formation of a spatial Si–O–Si network. Calcination conditions have a significant effect on the degree of structural order, specific surface area, and porosity of the final product. It has been found that under optimal temperature conditions, an amorphous mesoporous silicon dioxide is formed, possessing a high specific surface area, reduced content of surface hydroxyl groups, and increased thermal stability. The preservation of the amorphous state ensures pronounced potential applicability in adsorption and catalytic systems of the material. The developed technological approach, based on sequential carbonization, chemical demineralization, and thermal calcination, allows for control over the phase composition and morphology of the resulting product. It has been established that the formation of SiO_2 occurs due to the oxidative removal of the carbon matrix and the condensation of biogenic silica, rather than through the reduction of silicon.

Reduced energy consumption - the process works at moderate temperatures (700–1100 °C) instead of the high temperature (>1700 °C needed for carbothermal reduction of quartz);

Resource-efficient – SiO_2 yield of 30-35 wt.% relative to carbonized rice husks and with final purity >95%;

Higher textural properties – the produced silica has a superior quality, has a well-defined mesoporous structure, has a specific surface area of 120–150 m^2/g and a total pore volume of 0.175–0.287 cm^3/g;

Dual functionalisation - natural synthesis - using this method naturally creates a synthesized silicon oxides from the rice husk into SiO_2-C composite precursor that eliminates the need to introduce carbon addition in many applications;

Environmental advantages – the approach addresses the above and the problem of rice husk waste disposal, and substantially reducing CO_2 emissions and consuming less energy compared to conventional mining and high-temperature processing of quartz at rock and high-temperature processing. The obtained biogenic amorphous SiO_2 possesses structural characteristics that make it a promising precursor for electrochemical applications, especially in the presence of carbon matrix. The SiO_2–C composites are very suitable anode materials for lithium-ion batteries and lithium-ion capacitors, where a high theoretical capacity obtained from silicon, as well as low volume expansion due to carbon, is advantageous throughout charge–discharge cycles. By this process, we advance the principles of the circular economy and green chemistry by converting underused agricultural waste into high-potential silicon-based materials. Given the locally sourced raw materials, the technology is very applicable for Kazakhstan and rice-producing countries. Subsequent studies will include:

Quantitative electrochemical performance testing (cycling stability, rate capability, and full-cell configuration);

Overall, it has been demonstrated that regulating thermal treatment conditions is an effective tool for controlling the structure and functional properties of silicon- and carbon-containing materials, which expands the possibilities for their targeted application in adsorption, catalytic, and electrochemical systems. The use of rice husks as a secondary raw material confirms the environmental and resource efficiency of the proposed approach and is consistent with the principles of "green chemistry" and sustainable natural resource management.

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Conflict of Interest

The authors declare that there is no conflict of interest.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used ChatGPT (OpenAI) for language editing, improvement of clarity, and refinement of English wording. After using this tool, the authors carefully reviewed and edited the content as needed and take full responsibility for the content of the published article

Supporting Information

Not applicable.

CRediT Statement

Alibek Mutushev: Conceptualization, Methodology,

Investigation, Writing – Original draft. **Assiya Nuraly:** Investigation, Formal analysis, Data curation, Writing – Original draft. **Aknur Seisenova:** Investigation, Project administration, Supervision, Validation, Data curation. **Omirzak Kapizov:** Methodology, Investigation, Formal analysis. **Dauren Mukhanov:** Investigation, Visualization. **Zarina Inelova:** Conceptualization, Supervision, Writing – Review & editing. **Zura Yessimsiitova:** Validation, Resources, Writing – Review & editing. **Akzhunis Akimbayeva:** Investigation, Data curation. **Rysmagambetova Aina:** Supervision, Validation, Writing – Review & editing. **Aitzhan Mengtay:** Conceptualization, Writing – Review & editing.

References

[1] A. Mutushev, A. Kaya, M. Tulepov, Z. Kudyarova, D. Baiseitov, D. Mukhanov, Design and development of carbon–silicon-based air purification filters with antibacterial properties, *Processes*, 2025, **13**, 662, doi: 10.3390/pr13030662.

[2] M. K. Kazankapova, B. T. Yermagambet, A. B. Malgazhdarova, A. S. Akshekina, Z. M. Kassenova, N. S. Akimbekov, K. T. Tastambek, A. M. Malik, U. M. Kozhamuratova, K. B. Kaliyev, N. Bakhytuly, D. A. Nussipov, B. A. Kapsalyamov, Rice husk as a low-cost adsorbent for environmental applications, *ES Energy & Environment*, 2026, **32**, 2218, doi: 10.30919/ee2218.

[3] L. T. Canham, Silicon quantum wire array fabrication by electrochemical and chemical dissolution of wafers, *Applied Physics Letters*, 1990, **57**, 1046-1048, doi: 10.1063/1.103561.

[4] J. Safarian, G. Tranell, M. Tangstad, Processes for upgrading metallurgical grade silicon to solar grade silicon, *Energy Procedia*, 2012, **20**, 88-97, doi: 10.1016/j.egypro.2012.03.011.

[5] Z. Saule, S. Shynggyskhan, M. Suleimenova, S. Azat, Y. Sailaukhanuly, M. Tulepov, Z. Tauanov, Enhanced adsorption of mercury and methylene blue using silver and surface modified zeolite-NaX derived from rice husk, *Engineered Science*, 2024, **31**, 1241, doi: 10.30919/es1241.

[6] P. Hiremath, N. Naik, R. Bhat, Z. Guo, B. S. Maddodi, G. S. Sandeep, D. B. Narasimha, N. Nagraj, Investigating the mechanical properties, durability, and environmental impact of partial cement substitution with slag cement and rice husk ash for sustainable concrete production, *ES Food & Agroforestry*, 2024, **18**, 1267, doi: 10.30919/esfaf1267.

[7] S. N. Mahdi, D. V. Babu R, M. Ohno, B. S. Thomas, B. Nagaratnam, N. Nawaukkaratharnant, Investigating the durability characteristics of brick fired rice husk ash as sustainable material for fly ash based geopolymer concrete, *Engineered Science*, 2025, **36**, 1633, doi: 10.30919/es1633.

[8] W. D. P. Rengga, R. D. Kusumaningtyas, H. Prasetiawan, S. Amkha, Water retention and nitrogen release behavior of slow-release fertilizers coated with rice husk biochar and natural adhesives, *ES Energy & Environment*, 2026, **31**, 2124, doi: 10.30919/ee2124.

[9] W. Zhang, L. Yan, M. Li, R. Zhao, X. Yang, T. Ji, Z. Gu, Jun-Jie Yin, X. Gao, G. Nie, Deciphering the underlying mechanisms of oxidation-state dependent cytotoxicity of graphene oxide on mammalian cells, *Toxicology Letters*, 2015, **237**, 61-71, doi: 10.1016/j.toxlet.2015.05.021.

[10] A. Nuraly, A. Mutushev, A. Tuleibayeva, J. M. Gonzalez-Leal, Experimental research on optimizing carbon materials for filtration applications in medicine, *Carbon Trends*, 2024, **15**, 100338, doi: 10.1016/j.cartre.2024.100338.

[11] R. Liu, M. Liu, J. Chang, Large-scale synthesis of monodisperse SiC nanoparticles with adjustable size, stoichiometric ratio and properties by fluidized bed chemical vapor deposition, *Journal of Nanoparticle Research*, 2017, **19**, doi: 10.1007/s11051-016-3737-y.

[12] M. Dasog, L. F. Smith, T. K. Purkait, J. G. C. Veinot, Low temperature synthesis of silicon carbide nanomaterials using a solid-state method, *Chemical Communications*, 2013, **49**, 7004-7006, doi: 10.1039/c3cc43625j.

[13] A. Seisenova, A. Nuraly, D. Baiseitov, O. Kapizov, S. Oryngaliyeva, Z. Alimkulova, A. Mutushev, Comprehensive assessment of ash and slag waste for the synthesis of silicon-based functional materials, *Processes*, 2025, **13**, doi: 10.3390/pr13092722.

[14] R. Cheung, Silicon Carbide Microelectromechanical Systems for Harsh Environments, *Imperial College Press*, London, 2006, 181, ISBN: 9781860946240.

[15] J. Qiu, Y. Li, Y. Wang, Z. Zhao, Y. Zhou, Y. Wang, Synthesis of carbon-encapsulated nickel nanocrystals by arc-discharge of coal-based carbons in water, *Fuel*, 2004, **83**, 615-617, doi: 10.1016/j.fuel.2003.09.005.

[16] K. Navakul, C. Warakulwit, Pa-thai Yenchitsomanus, A. Panya, P. A. Lieberzeit, C. Sangma, A novel method for dengue virus detection and antibody screening using a graphene-polymer based electrochemical biosensor, *Nanomedicine: Nanotechnology, Biology and Medicine*, 2017, **13**, 549-557, doi: 10.1016/j.nano.2016.08.009.

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